

Autodesk® Scaleform®

Scaleform 4.4 AS3

Scaleform 4.4

ActionScript 3.0

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1.0

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Autodesk® Scaleform® 4.4

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AS3

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1 引言

Autodesk Scaleform® SDK™

Adobe® Flash®

Autodesk Scaleform

PC
(Adobe Creative Suite®)

```
Flash      Web      ,      (  IME
      )      Autodesk Scaleform
"      "(Extension)      ActionScript      Scaleform
      ,      scaleform.gfx.Extensions.enabled      true( )
```

```
scaleform.gfx.Extensions.enabled = true;
```

```
scaleform.gfx.*;
Extensions.enabled = true;
```

```

GFXSDK/Resources/AS3/CLIK      Flash      "      (Source)      ",
ActionScript 3.0      (      Flash Studio      : 'Edit'(      ) -> 'Preferences'(      ) -> 'ActionScript' ->
'ActionScript 3.0 Settings...' (ActionScript 3.0      ...) -> 'Source path'(Source
))      ,      FLA
      , Scaleform      ,

```

Flash

Flash, Scaleform, SDK

Autodesk API
Scaleform API

2

enabled

enabled:Boolean [read-write]

Scaleform 4.0.12

/

noInvisibleAdvance

noInvisibleAdvance:Boolean [read-write]

Scaleform 4.0.12

true, SWF
Flash (ActionScript
,) , true Scaleform Flash

getTopMostEntity()

```
public function getTopMostEntity(x :Number, y :Number, testAll:Boolean)  
:DisplayObject
```

Scaleform 4.0.13

(x, y) (stage) DisplayObject
似 (似 ,CLICK MOUSE_DOWN ROLL_OVER
)

MovieClip.hitTest , hitTest x y , getTopMostEntity
x y ,Extensions.getTopMostEntity(x, y).hitTest(x, y, true) == true

testAll :Boolean - (false),
(true) , getTopMostEntity true

x :Number, y :Number - Stage

[getMouseTopMostEntity](#)

getMouseTopMostEntity()

```
public function getTopMostEntity([testAll : Boolean],
    [mouseIndex : uint]) : DisplayObject
```

Scaleform 4.0.13

DisplayObject 似

getTopMostEntity, ,

```
testAll :Boolean - (false),
(true) , getTopMostEntity true
mouseIndex :Number -
```

[getTopMostEntity](#)

getMouseCursorType()

```
public function getMouseCursorType([mouseIndex : uint]) : String
```

Scaleform 4.0.13

flash.ui.Mouse.cursor

,

, MouseEvent.CURSOR_CHANGE

mouseIndex :Number -

, flash.ui.MouseCursor ,似 :

- flash.ui.MouseCursor.ARROW : String = "arrow"

- [setMouseCursorType](#)
[MouseEvent](#)

```
public function setMouseCursorType(cursor : String, [mouseIndex : uint]) : void
```

flash.ui.Mouse.cursor

, `MouseEvent.CURSOR_CHANGE`

,似

[getMouseCursorType](#)
[MouseEvent](#)

```
numControllers:uint    [read]
```

14

3 DisplayObjectEx

setRendererString

```
static public function setRendererString(o:DisplayObject, s:String)
```

Scaleform : 4.0.17

MovieClip 似 ActionScript

似:

```
import scaleform.gfx.*;

// m is a MovieClip on stage.

DisplayObjectEx.setRendererString(m, "Abc");
trace(DisplayObjectEx.getRendererString(m));
```

setRendererFloat

```
static public function setRendererFloat(o:DisplayObject, f:Number)
```

Scaleform : 4.0.17

MovieClip 似 ActionScript

似

```
import scaleform.gfx.*;

// m is a MovieClip on stage.
```

```
DisplayObjectEx.setRendererFloat(m, 17.1717);  
trace(DisplayObjectEx.getRendererFloat(m));
```

disableBatching

```
static public function disableBatching(o:DisplayObject, b:Boolean)
```

Scaleform : 4.0.17

似

```
import scaleform.gfx.*;  
  
// m is a MovieClip on stage.  
  
DisplayObjectEx.disableBatching(batchDisable, true);  
trace(DisplayObjectEx.isBatchingDisabled(batchDisable));
```

4 FocusEventEx

```
package scaleform.gfx
{
    import flash.events.FocusEvent;

    public final class FocusEventEx extends FocusEvent
    {
        public var controllerIdx : uint = 0;

        public function FocusEventEx(type:String) { super(type); }
    }
}
```

```

flash.events.FocusEvent
'controllerIdx'
Extensions.enabled = true
,
Scaleform
FocusEventEx
FocusEventEx
似
,
FocusEvent
似
```

```
import scaleform.gfx.*;
import flash.events.FocusEvent;

Extensions.enabled = true;

function ev(e: FocusEvent)
{
    if (e is FocusEventEx)
    {
        var ee: FocusEventEx = e as FocusEventEx;
        trace("    controllerIdx = "+ee.controllerIdx);
    }
}

stage.addEventListener(FocusEvent.MOUSE_FOCUS_CHANGE, ev);
```

controllerIdx

```
controllerIdx : uint [read]
```

Scaleform 4.0.12

/ ()

5 FocusManager

alwaysEnableArrowKeys

```
alwaysEnableArrowKeys:Boolean [read-write]
```

Scaleform 4.0.12

```

-- _focusrect
false( ) , _focusrect = false ,Flash
, alwaysEnableArrowKeys true
```

disableFocusKeys

```
disableFocusKeys:Boolean [read-write]
```

Scaleform 4.0.12

```
(TAB Shift-TAB ),
```

moveFocus()

```
static public function moveFocus(keyToSimulate : String,
    startFromMovie:InteractiveObject = null,
    includeFocusEnabledChars : Boolean = false,
    controllerIdx : uint = 0) : InteractiveObject
```

Scaleform 4.0.12

```

:TAB Shift-
TAB disableFocusKeys modalClip ,
```

```
keyToSimulate :String-
"shifttab"
" " " " " " " " " "tab"
```

```

startFromMovie:InteractiveObject - , ;moveFocus
( ) (Null)
, ;
includeFocusEnabledChars :Boolean - , moveFocus focusEnabled
tabEnabled / tabIndex
false, tabEnabled / tabIndex
controllerIdx :uint - , ( 0)

, null

```

[findFocus](#)
[disableFocusKeys](#)
[setModalClip](#)
[getModalClip](#)

findFocus()

```

static public function findFocus(keyToSimulate : String,
    parentMovie:DisplayObjectContainer = null,
    loop : Boolean = false,
    startFromMovie:InteractiveObject = null,
    includeFocusEnabledChars : Boolean = false,
    controllerIdx : uint = 0) : InteractiveObject

```

Scaleform 4.0.12

TAB disableFocusKeys setModalClip/getModalClip :TAB Shift-

```

keyToSimulate :String - " " " " " " " " " "tab" "shifttab"
parentMovie:DisplayObjectContainer - null
,
loop :Boolean - Boolean
似 , " " , findFocus "null"(
"false") ( "true")

```

```

startFromMovie:InteractiveObject - , ;moveFocus
( ) null
,
includeFocusEnabledChars :Boolean - , moveFocus focusEnabled
tabEnabled / tabIndex
false, tabEnabled / tabIndex
controllerIdx :uint -
,
null
,

```

[moveFocus](#)
[disableFocusKeys](#)
[setModalClip](#)
[getModalClip](#)

setFocus()

```

static public function setFocus(obj:InteractiveObject, controllerIdx:uint = 0) :void

```

Scaleform 4.0.12

stage.focus

,

```

obj:InteractiveObject -
controllerIdx :uint - / ( )

```

getFocus()

```

static public function getFocus(controllerIdx:uint = 0) :InteractiveObject

```

Scaleform 4.0.12

stage.focus ,

```

controllerIdx :uint - / ( )

```

numFocusGroups

numFocusGroups() :uint [read]

Scaleform 4.0.12

, setControllerFocusGroup 0 3
, numFocusGroups 2

setFocusGroupMask()

public function setFocusGroupMask(obj:InteractiveObject, mask:uint) :void

Scaleform 4.0.12

setControllerFocusGroup
,
似 , " 1" 0 , " 2" 0 1
,

```
FocusManager.setControllerFocusGroup(0, 0);  
FocusManager.setControllerFocusGroup(1, 1);
```

```
FocusManager.setFocusGroupMask(button1, 0x1); // 0 - 0  
FocusManager.setFocusGroupMask(movieclip2, 0x1 | 0x2); // 0 1 -  
// 0 1
```

"focusGroupMask"

obj:InteractiveObject -
mask:uint -

[setControllerFocusGroup](#)
[getFocusGroupMask](#)

getFocusGroupMask()

```
static public function getFocusGroupMask(obj:InteractiveObject) :uint
```

Scaleform 4.0.12

```
( , setFocusGroupMask)
```

obj:InteractiveObject -

[setControllerFocusGroup](#)

[setFocusGroupMask](#)

setControllerFocusGroup()

```
static public function setControllerFocusGroup(controllerIdx:uint,  
                                              focusGroupId:uint) : Boolean
```

Scaleform 4.0.12

```
controllerIndex , 0  
, , 似 ,  
( ), setControllerFocusGroup(1,1) 1  
setControllerFocusGroup(1,0) 0 1
```

controllerIdx:uint -

focusGroupId:uint -

, true

getControllerFocusGroup()

```
static public function getControllerFocusGroup(controllerIdx:uint) :uint
```

Scaleform 4.0.12

controllerIndex -

setModalClip()

```
static public function setModalClip(mc:Sprite, controllerIdx:uint = 0) : void
```

Scaleform 4.0.12

“tabable” “(modal) TAB Shift-TAB

controllerIdx:uint -
mc:Sprite -

getModalClip()

```
static public function getModalClip(controllerIdx:uint = 0) : Sprite
```

Scaleform 4.0.12

controllerIdx -

```

    (
        Scaleform
    )
    SF_KeyCodes.h
    SWF

    ,Scaleform FxPlayer

    ,
    Gfx::Movie::HandleEvent
    ,
    AS3
    ,
    Scaleform FxPlayer
    GamePadAnalogEvents
    GamePad

```

supportsAnalogEvents()

```
public static function supportsAnalogEvents() :Boolean
```

Scaleform 4.0.13

```
Scaleform (
true GamePadAnalogEvents
```

GamePadAnalogEvent

7 GamePadAnalogEvent

```
package scaleform.gfx
{
    import flash.events.Event;

    public final class GamePadAnalogEvent extends Event
    {
        public static const CHANGE:String = "gamePadAnalogChange";

        public var code :uint = 0; // scaleform.gfx.GamePad
        //
        public var controllerIdx : uint = 0;
        public var xvalue :Number = 0; // [-1, 1]
        public var yvalue :Number = 0; // [-1, 1]

        public function GamePadAnalogEvent(bubbles:Boolean, cancelable:Boolean,
                                           code:uint, controllerIdx:uint = 0,
                                           xvalue:Number = 0, yvalue:Number = 0)
        {
            super(GamePadAnalogEvent.CHANGE, bubbles, cancelable);
            this.code = code;
            this.controllerIdx = controllerIdx;
            this.xvalue = xvalue;
            this.yvalue = yvalue;
        }
    }
}
```

Scaleform , Stage
 , Stage Scaleform FxPlayer 'code' GamePad
 , , , Scaleform () 似:

```
import scaleform.gfx.*;

trace("GamePadAnalogEvents supported? " + GamePad.supportsAnalogEvents());

function ev(e: GamePadAnalogEvent)
{
    trace("code = " + ev.code);
    trace("controllerIdx = " + ev.controllerIdx);
    trace("xvalue = " + ev.xvalue);
    trace("yvalue = " + ev.yvalue);
}
```

```

}
stage.addEventListener(GamePadAnalogEvent.CHANGE, ev);

```

code

code : uint

Scaleform 4.0.13

/

Scaleform FxPlayer

[GamePad](#)

controllerIdx

controllerIdx : uint

Scaleform 4.0.13

/

()

xvalue

xvalue : Number

Scaleform 4.0.13

x

-1 1

,

-1 1

yvalue

yvalue : Number

Scaleform 4.0.13

y

-1 1

,

-1 1

8 InteractiveObjectEx

getHitTestDisable()

```
public function getHitTestDisable(o:InteractiveObject) : Boolean
```

Scaleform 4.0.13

```
'hitTestDisable' true ,MovieClip.hitTest
,
false
```

o - An interactive object.

```
'hitTestDisable' Boolean
```

[InteractiveObjectEx.setHitTestDisable](#)

setHitTestDisable()

```
public function setHitTestDisable(o:InteractiveObject, f:Boolean) : void
```

Scaleform 4.0.13

```
'hitTestDisable' true ,MovieClip.hitTest
, false
```

o -

f - 'hitTestDisable' Boolean

[InteractiveObjectEx.getHitTestDisable](#)

getTopmostLevel()

```
public function getTopmostLevel(o:InteractiveObject) : Boolean
```

Scaleform 4.0.13

'topmostLevel' true , ,

o - An interactive object.

'topmostLevel' Boolean

[InteractiveObjectEx.setTopmostLevel](#)

setTopmostLevel ()

public function setTopmostLevel(o:InteractiveObject, f:Boolean) : void

Scaleform 4.0.13

```
'topmostLevel' true
,
, false
"topmostLevel",
, A B "topmostLevel" B true
: "topmostLevel",swapDepth ActionScript
false
```

o -

f - 'topmostLevel' Boolean

[InteractiveObjectEx.getTopmostLevel](#)

9 KeyboardEventEx

```
package scaleform.gfx
{
    import flash.events.KeyboardEvent;

    public final class KeyboardEventEx extends KeyboardEvent
    {
        public var controllerIdx : uint = 0;

        public function KeyboardEventEx(type:String) { super(type); }
    }
}
```

```

        flash.events.KeyboardEvent
        ,
        ,Scaleform
KeyboardEvent
        似,
        KeyboardEventEx
        ,
        KeyboardEventEx
        似
        'controllerIdx'
        Extensions.enabled
        true
```

```
import scaleform.gfx.*;
import flash.events.KeyboardEvent;

Extensions.enabled = true;

function ev(e: KeyboardEvent)
{
    if (e is KeyboardEventEx)
    {
        var ee: KeyboardEventEx = e as KeyboardEventEx;
        trace("    controllerIdx = "+ee.controllerIdx);
    }
}

stage.addEventListener(KeyboardEvent.KEY_DOWN, ev);
stage.addEventListener(KeyboardEvent.KEY_UP, ev);
```

controllerIdx

```
controllerIdx : uint [read]
Scaleform 4.0.12
/ ( )
```


10 MouseCursorEvent

```
package scaleform.gfx
{
    import flash.events.Event;

    public final class MouseCursorEvent extends Event
    {
        public var cursor : String = "auto";
        public var mouseIdx : uint = 0;

        static public const CURSOR_CHANGE : String = "mouseCursorChange";

        public function MouseCursorEvent()
        {
            super("MouseCursorEvent", false, true);
        }
    }
}

/ : bubbles – false, (bubble)
cancellable – true, preventDefault() ( )
Scaleform
( , flash.ui.Mouse.cursor ), stage
( Extensions.enabled true ) 似:

Extensions.enabled = true;

function e(e:MouseCursorEvent)
{
    trace(e.type + " " + e.mouseIdx + " " + e.cursor);
    e.preventDefault();
}
stage.addEventListener(MouseCursorEvent.CURSOR_CHANGE, e);

: Stage ,
```

cursor

cursor : String [read]

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- `flash.ui.MouseCursor.ARROW` : String = "arrow"
- `flash.ui.MouseCursor.BUTTON` : String = "button"
- `flash.ui.MouseCursor.HAND` : String = "hand"
- `flash.ui.MouseCursor.IBEAM` : String = "ibeam"

mouseIdx

`mouseIdx` : uint [read]

Scaleform 4.0.13

/ ()

11 MouseEventEx

```
package scaleform.gfx
{
    import flash.events.MouseEvent;

    public final class MouseEventEx extends MouseEvent
    {
        public var mouseIdx : uint = 0;
        public var nestingIdx : uint = 0;
        public var buttonIdx : uint = 0; // LEFT_BUTTON, RIGHT_BUTTON, ...

        public static const LEFT_BUTTON : uint = 0;
        public static const RIGHT_BUTTON : uint = 1;
        public static const MIDDLE_BUTTON : uint = 2;

        public function MouseEventEx(type:String) { super(type); }
    }
}
```

```

flash.events.MouseEvent
Extensions.enabled      true ,Scaleform      MouseEventEx      /
MouseEvent              MouseEventEx
    似,      ,      似:
import scaleform.gfx.*;
Extensions.enabled = true;

stage.doubleClickEnabled = true;

function ev(e:MouseEvent):void
{
    trace("!!!! EVENT. " + cnt++);
    trace("    eventType      = "+e.type);
    trace("    bubbles        = "+e.bubbles);
    trace("    eventPhase      = "+e.eventPhase);
    trace("    target            = "+e.target.name);
    trace("    currentTarget = "+e.currentTarget.name);
    if (e is MouseEventEx)
    {
        var ee:MouseEventEx = e as MouseEventEx;
        trace("    mouseIdx        = "+ee.mouseIdx);
        trace("    nestingIdx      = "+ee.nestingIdx);
        trace("    buttonIdx       = "+ee.buttonIdx);
    }
}
```

```

:
,
,
buttonIdx

```

```
buttonIdx : uint [read]
```

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Scaleform 4.0.13

- MouseEventEx.LEFT_BUTTON :uint = 0 -
- MouseEventEx.RIGHT_BUTTON :uint = 1 -
- MouseEventEx.MIDDLE_BUTTON :uint = 2 -
- 3 , 2

```
mouseIdx : uint [read]
```

Scaleform 4.0.13

Scaleform 4.0.13

nestingIdx

Scaleform 4.0.13

Scaleform 4.0.13

rollOver/Out mouseOver/Out dragOver/Out
/ (rollover/dragover)

```

rollOver/rollOut mouseOver/Out dragOver/dragOut
(
),
:
0
;
,
1,
rollOver/rollOut
rollOut/mouseOut/dragOut;
0,
rollOut/mouseOut/dragOut

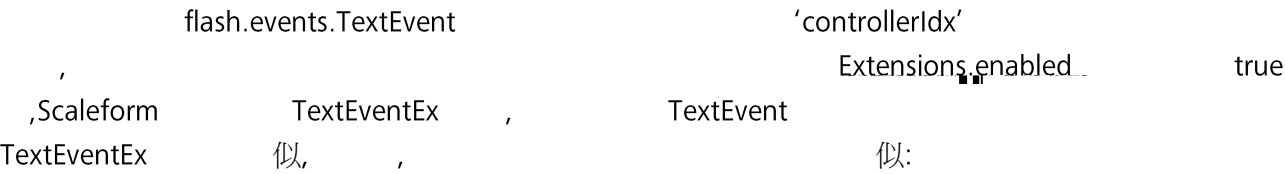
```

12 TextEventEx

```
package scaleform.gfx
{
    import flash.events.TextEvent;

    public final class TextEventEx extends TextEvent
    {
        public var controllerIdx : uint = 0;

        public function TextEventEx(type:String) { super(type); }
    }
}
```



```
import scaleform.gfx.*;
import flash.events.TextEvent;

Extensions.enabled = true;

function ev(e: TextEvent)
{
    if (e is TextEventEx)
    {
        var ee: TextEventEx = e as TextEventEx;
        trace("    controllerIdx = "+ee.controllerIdx);
    }
}

txf.addEventListener(TextEvent.TEXT_INPUT, ev);
```

controllerIdx

```
controllerIdx : uint [read]
```

Scaleform 4.0.12

/ ()

LINK_MOUSE_OVER/LINK_MOUSE_OUT

```
public static const LINK_MOUSE_OVER:String = "linkMouseOver";
public static const LINK_MOUSE_OUT:String = "linkMouse";
```

Scaleform 4.0.14

	TextFieldEx.LINK_MOUSE_OVER	TextFieldEx.LINK_MOUSE_OUT
TextField (HTML)	over/out	AS3
,	addEventListener	似

13 System

actionVerbose

`actionVerbose:Boolean` [read-write]

Scaleform 4.0.12

/

getStackTrace()

`public function getStackTrace() : String`

Scaleform 4.0.12

getCodeFileName()

`public function getCodeFileName() : String`

Scaleform 4.0.12

14 TextFieldEx

appendHtml()

```
public function appendHtml(textField:TextField, newHtml:String) :void
```

Scaleform 4.0.12

```
newHtml HTML
(+=)( txt.htmlText += moreHtml) htmlText += HTML , HTML
, HTML HTML , newHtml
HTML( newHtml HTML , +=
)
```

```
textField:TextField- HTML
newHtml:String- HTML
```

setIMEEnabled()

```
static public function setIMEEnabled(textField:TextField, isEnabled:Boolean): void
```

Scaleform 4.0.12

```
/ IME isEnabled false,
IME , IME
```

```
textField :TextField-
isEnabled :Boolean- true- IME;
```

setVerticalAlign()

```
public function setVerticalAlign(textField:TextField, valign:String) :void
```

TextFieldEx

```
public static const VALIGN_TOP:String      = "top";
public static const VALIGN_CENTER:String   = "center";
public static const VALIGN_BOTTOM:String   = "bottom";
```

```
        center,
bottom, ( ) ;
```



top

```
textField:TextField-
valign:String      - ("top", "center", "bottom")
```